

# Labor Demand

Raffaele Saggio

UBC

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- Theory of the firm  $\rightarrow$  Theory of labor demand  $\rightarrow$  Theory of Inequality
  - Demand for inputs (e.g. labor, capital) from optimization.
  - Key question: what explains the elasticity of demand for labor?
- Today: neo-classical labor demand.
- Building block for a variety of “applied” questions:
  - Effects of technology
  - Effects of immigration / demographics.
  - Effects of trade
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  - Some of these will populate the “Inequality” wars chapter.

# The building block

- Firms produce an output combining multiple inputs:

$$y = f(x_1, x_2, \dots, x_N) \quad (1)$$

- Typical assumptions:

- $f(\cdot)$  exhibits constant return to scale (CRS).
- work with only two inputs:  $y = f(x_1, x_2)$ .

- Notation time-out

- $f_i \equiv$  partial derivative of  $f$  with respect to  $x_i$  with  $i \in \{1, 2\}$ .

# An Old friend

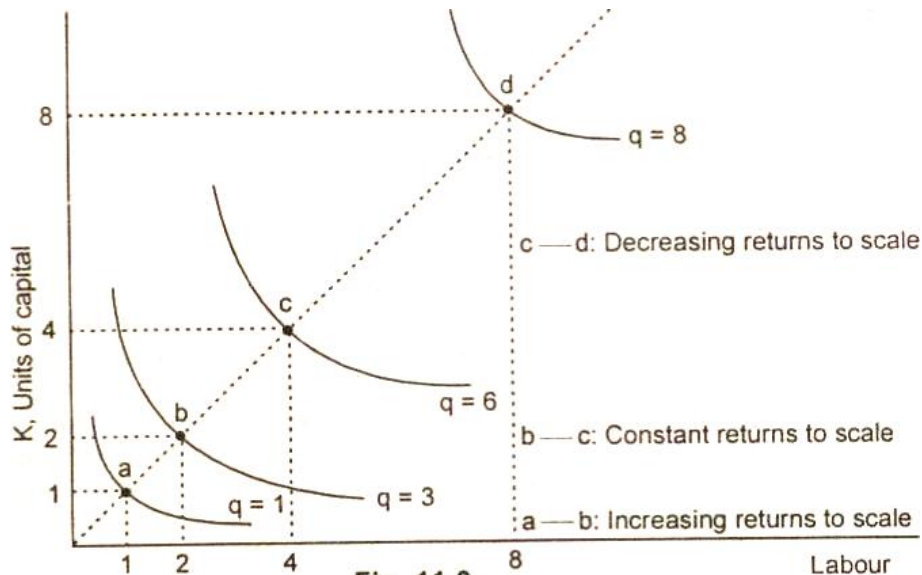


Fig. 11.6

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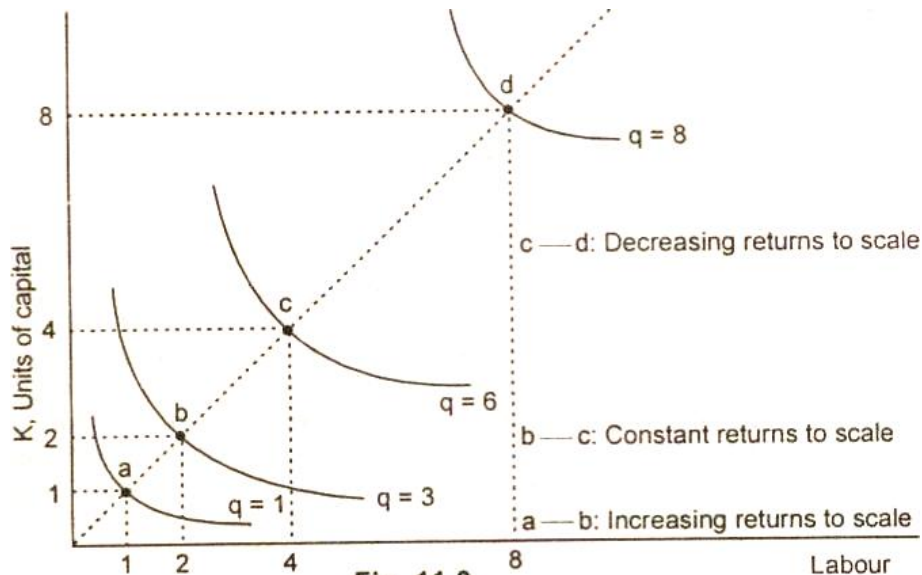


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# The Holy Grail: Elasticity of substitution

- Under CRS and two inputs: shape of the isoquant is going to be given by

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  - 2 Under perfect competition, what is this denominator?
- Name of the game in labor demand (most of the time): estimate  $\sigma$

# Special Cases

- Cobb-Douglas

$$y = f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha} \quad (3)$$

$$\sigma = 1$$

- CES:

$$y = f(x_1, x_2) = (\alpha x_1^{-\rho} + (1 - \alpha)x_2^{-\rho})^{-\frac{1}{\rho}} \quad (4)$$

$$\sigma = \frac{1}{(1 + \rho)}$$

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- Sheppard's Lemma (Envelope Theorem):

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- Important: all derivations that are coming up are for a fixed output  $y$ .

- We have that  $C_i$  is homogenous of degree 0 ( $HD_0$ ) , then

$$x_1 = C_1\left(\frac{w_1}{w_2}, 1, y\right) \equiv g\left(\frac{w_1}{w_2}, y\right) \quad (8)$$

$$x_2 = C_2\left(\frac{w_1}{w_2}, y\right) \equiv h\left(\frac{w_1}{w_2}, y\right) \quad (9)$$

- Then we have that

$$C_{11} = \frac{\partial x_1}{\partial w_1} = \frac{1}{w_2} g_1\left(\frac{w_1}{w_2}, y\right) \quad (10)$$

$$C_{21} = \frac{\partial x_2}{\partial w_1} = \frac{1}{w_2} h_1\left(\frac{w_1}{w_2}, y\right) \quad (11)$$



# General Case

- Then, for a fixed  $y$ ,

$$\log \frac{x_1}{x_2} = \log g\left(\frac{w_1}{w_2}, y\right) - \log h\left(\frac{w_1}{w_2}, y\right) \quad (12)$$

- Putting the pieces together

$$\begin{aligned} \sigma &= - \frac{\left(\frac{w_1}{w_2}\right) d \log\left(\frac{x_1}{x_2}\right)}{d \frac{w_1}{w_2}} = - \frac{w_1}{w_2} \left( \frac{g_1\left(\frac{w_1}{w_2}, y\right)}{g} - \frac{h_1\left(\frac{w_1}{w_2}, y\right)}{h} \right) \\ &= - \frac{w_1}{w_2} \left( \frac{w_2 C_{11}}{g} - \frac{C_{21} w_2}{h} \right) \\ &\equiv - \frac{w_1 C_{11}}{C_1} + \frac{w_1 C_{21}}{C_2} \end{aligned} \quad (13)$$

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$$C_{11}w_1 + C_{12}w_2 = 0 \quad (14)$$

- Based on the above, we arrive at

$$\sigma = \frac{w_1 C_{21}}{C_2} + \frac{w_2 C_{12}}{C_1} = C_{12} \frac{(w_1 C_1 + w_2 C_2)}{C_1 C_2} \quad (15)$$

- Applying Euler's theorem to  $C(w_1, w_2)$  (which recall is a function  $HD_1$ ), that is

$$w_1 C_1 + w_2 C_2 = C(w_1, w_2) \quad (16)$$

- We finally obtain

$$\sigma = \frac{C_{12} C}{C_1 C_2} \quad (17)$$

- Extension: partial elasticity of substitution multi-factor production functions

$$\sigma_{ij} = \frac{C_{ij} C}{C_i C_j} \quad (18)$$

# Connecting the dots..

- Why these derivations are useful? Because we can start to see the importance of  $\sigma$  in driving quantities that we care about...

$$\begin{aligned}\frac{\partial x_1}{\partial w_2} &= C_{12} \\ \rightarrow \epsilon_{12} &\equiv \frac{w_2}{x_1} \frac{\partial x_1}{\partial w_2} = \frac{w_2 C_{12}}{x_1} \\ &= \frac{C_{12} C}{C_1 C_2} \frac{w_2 C_2 C_1}{C x_1} \\ &= \sigma \frac{w_2 x_2}{C} \\ &= \sigma \underbrace{s_2}_{\text{Input 2 cost share}}\end{aligned}\tag{19}$$

- Also, we have that

$$\begin{aligned}w_1 \frac{\partial x_1}{\partial w_1} + w_2 \frac{\partial x_1}{\partial w_2} &= 0 \\ \rightarrow \epsilon_{11} &\equiv \frac{\partial x_1}{\partial w_1} \frac{w_1}{x_1} = -\sigma s_2 = -(1 - s_1)\sigma\end{aligned}\tag{20}$$

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- Important reminder: we were holding  $y$  fixed thus far!

# Marshall's Laws

- We are now ready to derive Marshall's Laws for labor demand.
- These laws are derived under the following setup:
  - Perfectly competitive industry with CRS and cost function  $C(w, y) = \mu(w)y$  where  $w = (w_1, \dots, w_N)$ .
  - Output is priced at  $p = \mu(w)$ .
  - Downward sloping demand curve for the industry's output  $y = D(p)$ .
- Next, we will show how the elasticity of demand for an input  $x_i$  relates to three key parameters
  - 1  $\eta \equiv$  elasticity of product demand.
  - 2  $\sigma_{ij} \equiv$  partial elasticity of substitution.
  - 3  $s_j \equiv$  cost shares.

# Marshall's Laws

$$x_i = y\mu_i(w)$$

$$\rightarrow \log x_i = \log y + \log \mu_i(w).$$

$$\rightarrow \frac{\partial \log x_i}{\partial w_j} = \frac{\partial \log y}{\partial \log p} \frac{\partial \log p}{\partial w_j} + \frac{\mu_{ij}(w)}{\mu_i(w)}$$

$$= -\eta \frac{\partial \log \mu(w)}{\partial w_j} + \frac{\mu_{ij}}{\mu_i}$$

$$= -\eta \frac{\mu_j}{\mu} + \frac{\mu_{ij}}{\mu_i}$$

$$= -\eta \frac{\mu_j}{\mu} + \frac{\mu_{ij}\mu}{\mu_i\mu_j} \frac{\mu_j}{\mu}$$

$$= -\eta \frac{\mu_j}{\mu} + \sigma_{ij} \frac{\mu_j}{\mu}$$

$$= (\sigma_{ij} - \eta) \frac{\mu_j}{\mu} = (\sigma_{ij} - \eta) \frac{x_j}{\mu} \rightarrow w_j \frac{\partial \log x_i}{\partial w_j} = (\sigma_{ij} - \eta) \underbrace{\frac{w_j x_j}{\mu}}_{s_j} = \epsilon_{ij} - \eta s_j$$

# Marshall's Laws

- Key relationship:

$$\frac{\partial \log x_i}{\partial \log w_j} = \underbrace{\epsilon_{ij}}_{\text{Substitution Effect}} - \underbrace{\eta s_j}_{\text{Scale Effect}} \quad (22)$$

- Substitution Effect: how much does demand for input  $x_i$  increases for an increase in the price of input  $x_j$ , *holding  $y$  fixed*.
- Scale Effect: to understand this, note that since  $\mu(w) = p$

$$\begin{aligned} \frac{\partial \log p}{\partial w_j} &= \frac{\mu_j(w)}{\mu(w)} \\ \frac{w_j \partial \log p}{\partial w_j} &= \frac{w_j \mu_j(w) y}{\mu(w) y} = \frac{w_j x_j}{C} = s_j \end{aligned} \quad (23)$$

Thus, when  $w_j$  increases by 1%, industry selling price rises by  $s_j$  and this chokes off demand by  $\eta s_j$ %



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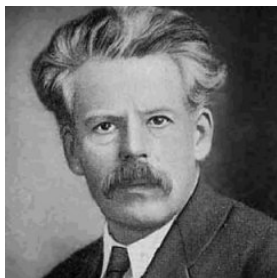
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- Does this bring back any memories?



# Discussion

- Neo-classical labor demand framework  $\rightarrow$  national or industry wide changes to the wage structure.
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- Neo-classical labor demand framework  $\rightarrow$  national or industry wide changes to the wage structure.
- Firms: all have CRS technology and are price-takers in the input market.
- Number and size of the firms is indeterminant.
- Very different from other fields!
- IO, Trade: lots of firm level heterogeneity in productivity. Does this productivity heterogeneity “spills” into wages?
- Topic of the last two lectures.

# Getting ready for war

- Useful benchmark is the Cobb-Douglas CRS benchmark ( $\sigma = 1$ )

$$y = f(K, L) = AL^\alpha K^{(1-\alpha)} \quad (24)$$

- Note that:

$$\begin{aligned} w &= \alpha AL^{\alpha-1} K^{1-\alpha} \\ wL &= \alpha AL^\alpha K^{1-\alpha} \\ \frac{wL}{y} &= \alpha \end{aligned} \quad (25)$$

- $\alpha$  is the labor share (how much of GDP goes to labor?).
  - An entire generation of economists thought that  $\alpha = 0.66$ .
  - Actually declining over the last years (active area of research).
- Starting point to understand productivity trends:

$$\underbrace{\frac{y}{L}}_{\text{Average Product}} = A \left( \frac{K}{L} \right)^{1-\alpha} \quad (26)$$

- Therefore in the C-D marginal product is just average product times a constant

$$w = \frac{\partial y}{\partial L} = \alpha \frac{y}{L} \quad (27)$$

# Labor Share is declining in many places...

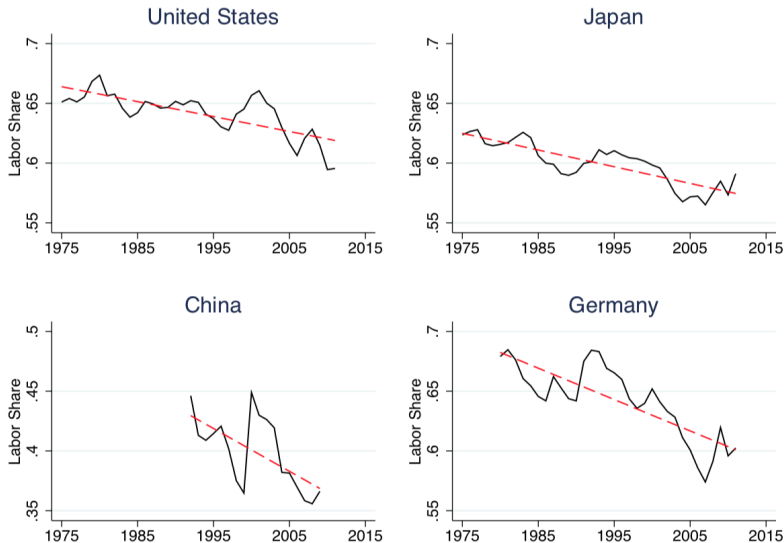


Figure 2: Declining Labor Share for the Largest Countries

# Something deep is happening now....

## The gap between productivity and a typical worker's compensation has increased dramatically since 1973

Productivity growth and hourly compensation growth, 1948–2017

